

Software Defined HPC with Open OnDemand

Global Open OnDemand 2025

Boston, Massachusetts

March 19, 2025

John-Paul Robinson, Louis Chen, Eesaan Atluri, Krish Moodbidri

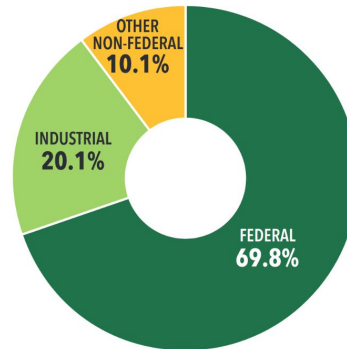
Research Computing Development Team

The University of Alabama at Birmingham (UAB)

About University of Alabama at Birmingham

- Urban research university located in Birmingham Alabama
- Ranked in top 5% of Public US Universities in Federal R&D Expenditures
- Ranked 13th among Public US Universities in NIH Awards
- Enrollment Fall 2024 20,905:
 - 11,595 Undergraduate, 5,331 Graduate, 3,615 Doctorate
- Faculty 3,265
- Professional staff 13,234 (includes hospital)

TOTAL GRANTS AWARDED



UAB ANNUAL RESEARCH EXPENDITURES

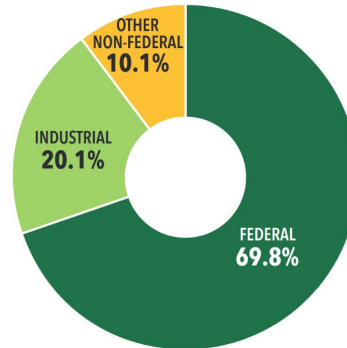


About University of Alabama at Birmingham



- Urban research university located in Birmingham Alabama
- Ranked in top 5% of Public US Universities in Federal R&D Expenditures
- Ranked 13th among Public US Universities in NIH Awards
- Enrollment Fall 2024 20,905:
 - 11,595 Undergraduate, 5,331 Graduate, 3,615 Doctorate
- Faculty 3,265
- Professional staff 13,234 (includes hospital)

TOTAL GRANTS AWARDED



UAB ANNUAL RESEARCH EXPENDITURES



Research Computing at UAB

- Research Computing is a dedicated unit of campus Information Technology
 - Science Engagement Team
 - Developers
 - Operations
- Provides compute and storage for campus research community
- Cheaha High Performance Computing (HPC) cluster, largest in Alabama
 - 2x 100 Intel & AMD nodes with 48 to 128 cores
 - RAM 512G and 768G with a handful of 1.5TB
 - 100G Infiniband Network
 - 2PB GPFS parallel file system
 - 100's of active users
- cloud.rc OpenStack platform for ad-hoc compute and dev
- 20+PB Ceph Storage Platforms (Block, S3, CephFS)
- On-campus and commercial data center connected by dark fiber at 800Gbps

Research Computing System (RCS)

Compute

HPC – Slurm Batch

VM – OpenStack Cloud

Container – Kubernetes

Storage

File System – GPFS & CephFS

Block – Ceph RBD

Object – Ceph RadosGW (S3)

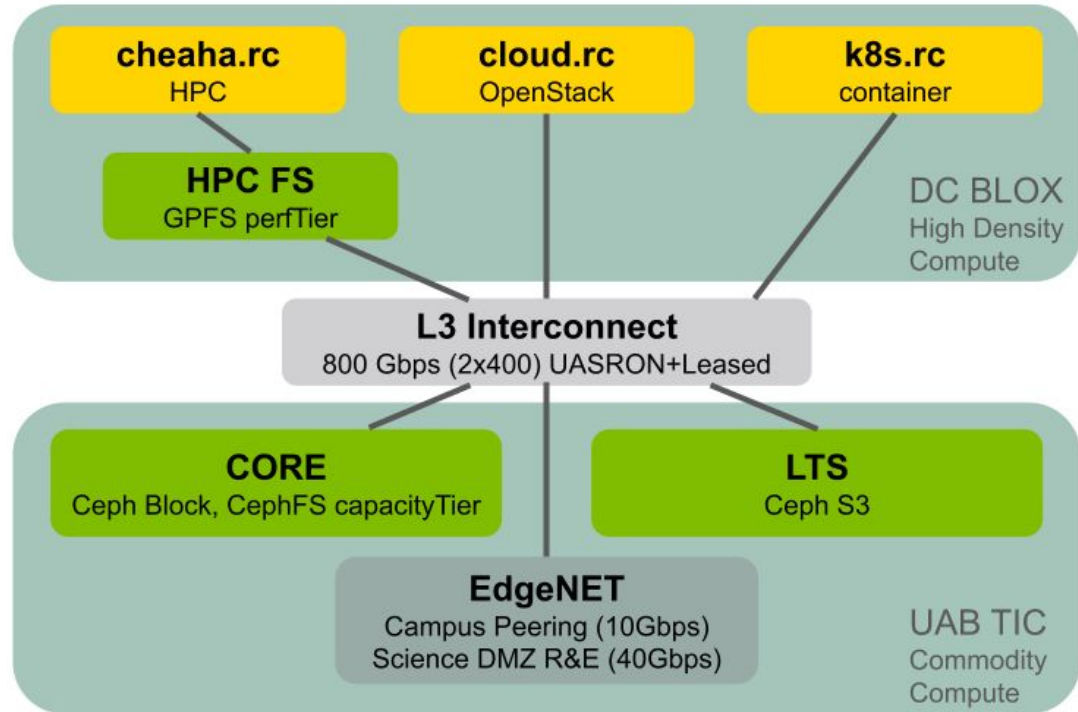
Network

Ethernet – 200G East/West

Interconnect – 800G

Infiniband – HDR and EDR

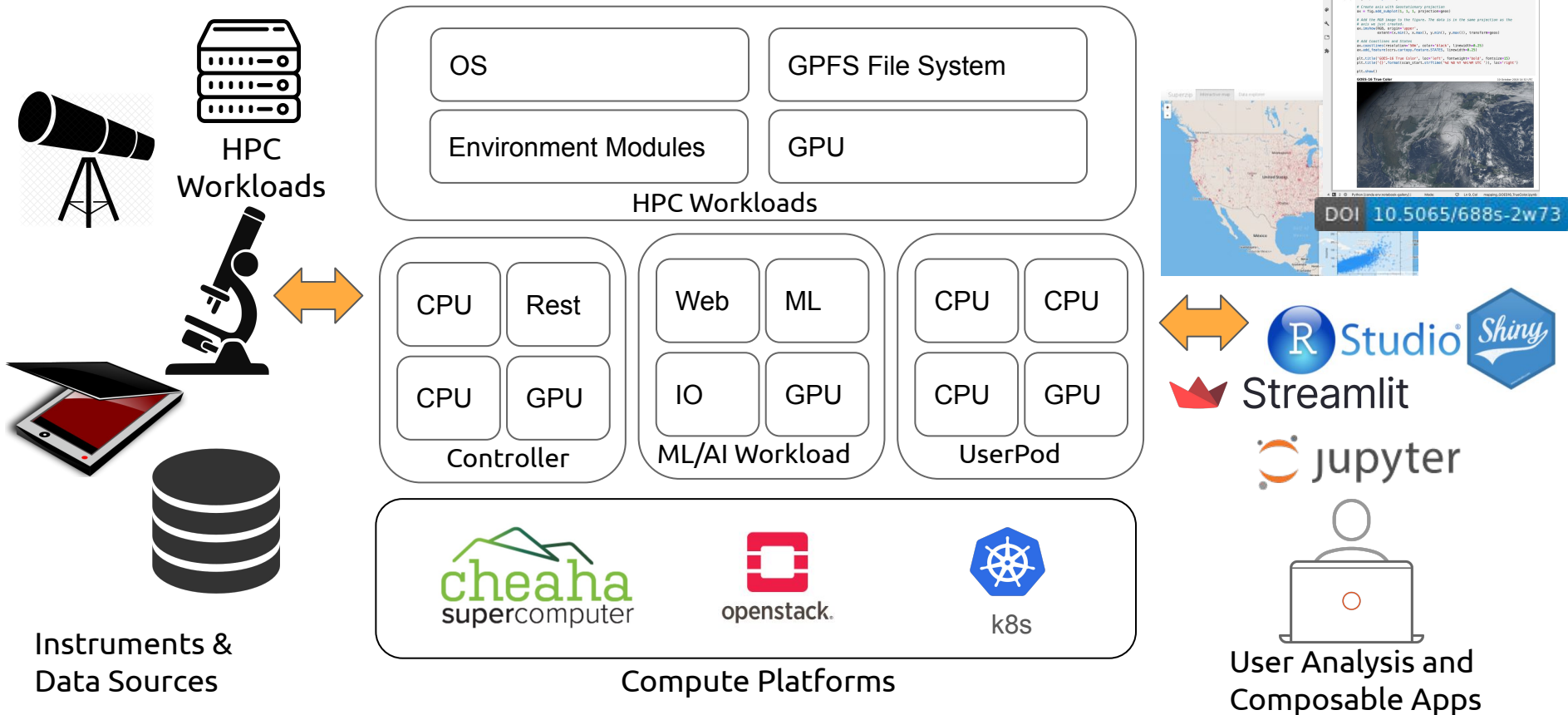
ScienceDMZ – 40G (SciDMZ)



Collaboration and Operations Tooling

- Open OnDemand (OOD) – web interface to HPC
- GitLab (or GitHub) – software development
- Science Gateways – XNAT for Nueroimaging
- Logging, Monitoring and Alerting – Graphana, XDMod
- APIs – enable tool building
- Documentation – enable self service

Preparing for NextGen Science Gateways



how did we get here?

Deployed Open OnDemand in May 2019

Had been looking for the “right” HPC portal...

Found Open OnDemand and fell in love. ❤️

Deployed as a “second” login node to our cluster with BCM

A great success story!

UAB THE UNIVERSITY OF ALABAMA AT BIRMINGHAM

UAB Reporter
News and information by and for the UAB community

Q

☰

Resources

Your browser is the supercomputer: On Demand is a no-tears shortcut to research-computing

Matt Windsor | UAB Reporter

June 24, 2019 | Print | Email

Here is the old way for new users to access Cheaha, UAB's supercomputer:

- Sign up for a Cheaha account.
- Open a Terminal window.
- Type in a list of commands.
- Send an email to Research Computing about the errors you get.

The traditional method of logging on to the Cheaha cluster, the fastest in Alabama, was “like something out of ‘Tron’ or ‘War Games,’” said William Monroe, the UAB IT Research Computing scientist who often was on the other end of those emails.



John-Paul Robinson, high-performance computing architect, displays On Demand in front of the Cheaha supercomputer.

Today there is UAB Research Computing On Demand, a web portal — at rc.uab.edu — that brings Cheaha supercomputer access as close as the nearest web browser. “This helps people who are not ‘command-line commandos,’ who are used to a windowed interface,” said Ralph Zottola, Ph.D., assistant vice president for research computing. ([Learn more about Zottola and his plans for UAB Research Computing in this related story.](#)) “It’s opening access to a lot of new users. And we’ve already seen a benefit in

The New HPC

New logo (IT marketing)

New docs (Science Team)

Account Self Registration
via SSO (Dev Team)

“Onboard to HPC over a
cup of coffee from the
comfort of your web
browser”

The collage displays four browser windows showcasing the UAB Research Computing platform:

- Top Left:** The main dashboard at rc.uab.edu/pun/sys/. It features the 'cheaha supercomputer' logo and a sidebar menu with categories: Desktops (HPC Desktop), GUIs (ANSYS, IGV, MATLAB, SAS), and Servers (Jupyter Notebook, JupyterLab, RStudio Server). The main content area includes a 'Message of the Day' and a table of available Slurm partitions.
- Top Right:** An 'Account in Good Standing' notification from the dashboard, stating: 'Your account is in good standing. Click [here](#) to proceed to dashboard.'
- Bottom Left:** A table titled 'Available Slurm Partitions' showing different resource configurations.
- Bottom Right:** The 'Research Computing Documentation' page, which provides information about the HPC fabric, data management, and contact details.

Partition	Nodes	Nodes Per Researcher	Time Limit
CPU			
interactive	52	1	2 hours
express	52	UNLIMITED	2 hours
short	52	44	12 hours
medium	52	44	2 days, 2 hours

then things got busy...

Life in UAB RC since 2019

Two months after the article came out the first 936 event happened Aug 1 2019

Fall 2019 JBOD Ceph hardware build 3PB

Jan 2020 set a Sprint Schedule

March 2020 went home for a while

May 2020 CPU compute expansion

June 2020 the second data center event event

July 2020 core HPC switch failure

October 2020 expansion to DC BLOX (DCB) and build of Openstack with DGX A100

May 2021 complete first Canonical engagement

July 2021 Plan k8s purchase and 34kW racks build in DC BLOX

Fall 2021 moved out of old data center into TIC + DCB with 34kW rack expansion

Spring 2022 K8s release

Fall 2022 kubecon, globus S3 connector

Life in UAB RC since 2019

Jan 2023 deploy Canonical Ceph Core cluster

Summer 2023 Campus Strategic Planning

Need an HPC storage solution because our original vendors broke up

OK, consolidate compute in DCB DH2 for RC

Dec 2023 order GPFS5 and L3 Interconnect

April 2024 finish GPFS5 deploy

May 2024 kick off migration planning

July 2024 select software, plan move

Oct 1 2024 migrate to new BGP core via dell resident

Jan 2024 we gotta move!

Life in UAB RC since 2019

Jan 2023 deploy Canonical Ceph Core cluster

Summer 2023 Campus Strategic Planning

Need an HPC storage solution because our original vendors broke up

OK, consolidate compute in DCB DH2 for RC

Dec 2023 order GPFS5 and L3 Interconnect

April 2024 finish GPFS5 deploy

May 2024 kick off migration planning

July 2024 select software, plan move

Oct 1 2024 migrate to new BGP core via dell resident

Jan 2024 we gotta move!

...ugh, we forgot to upgrade Open OnDemand. 😭

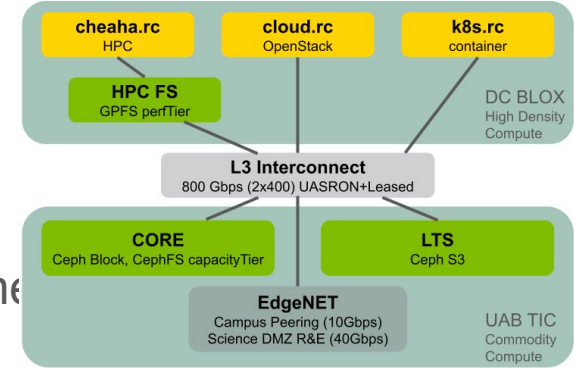
what to do? what to do?

Project: Re-organize Infrastructure across Data Centers

- Upgrade GPFS4 to GPFS5
 - Not in-place
 - Change hardware vendors
 - Change to tiered architecture to reduce cost 7PB -> 2PB
 - Power constrained in current campus HPC compute
- Consolidate Compute in power-dense racks of commercial facility
 - HPC and GPU infrastructure is demanding more power
 - Rear Door Heat Exchanger support 34kW 48U racks
 - Great for our Compute
- Consolidate Storage in low-power on-campus enterprise data center
 - Ceph provides all storage modalities for block, object and file
 - Provide scalable backend for growing research data sets
- Upgrade dark-fiber L2 interconnect to 2x400G with BGP Core
 - Ample capacity to move data between on campus and off campus facilities

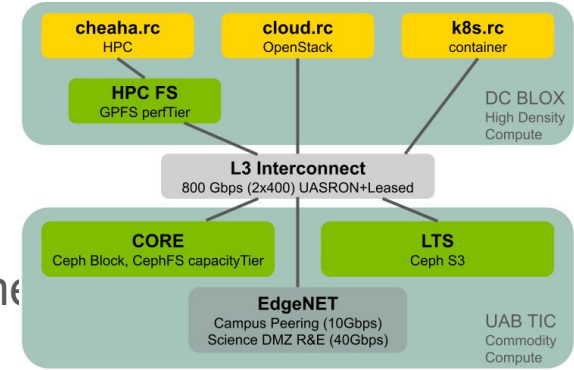
Project: Re-organize Infrastructure across Data Centers

- Upgrade GPFS4 to GPFS5
 - Not in-place
 - Change hardware vendors
 - Change to tiered architecture to reduce cost 7PB -> 2PB
 - Power constrained in current campus HPC compute
- Consolidate Compute in power-dense racks of commerce
 - HPC and GPU infrastructure is demanding more power
 - Rear Door Heat Exchanger support 34kW 48U racks
 - Great for our Compute
- Consolidate Storage in low-power on-campus enterprise data center
 - Ceph provides all storage modalities for block, object and file
 - Provide scalable backend for growing research data sets
- Upgrade dark-fiber L2 interconnect to 2x400G with BGP Core
 - Ample capacity to move data between on campus and off campus facilities



Project: Re-organize Infrastructure across Data Centers

- Upgrade GPFS4 to GPFS5
 - Not in-place
 - Change hardware vendors
 - Change to tiered architecture to reduce cost 7PB -> 2PB
 - Power constrained in current campus HPC compute
- Consolidate Compute in power-dense racks of commerce
 - HPC and GPU infrastructure is demanding more power
 - Rear Door Heat Exchanger support 34kW 48U racks
 - Great for our Compute
- Consolidate Storage in low-power on-campus enterprise data center
 - Ceph provides all storage modalities for block, object and file
 - Provide scalable backend for growing research data sets
- Upgrade dark-fiber L2 interconnect to 2x400G with BGP Core
 - Ample capacity to move data between on campus and off campus facilities



...oh and let's minimize user impact as much as possible

btw, this is a great time to demonstrate the
advantages of a dynamically provisioned
infrastructure

Approach for moving a live HPC system...

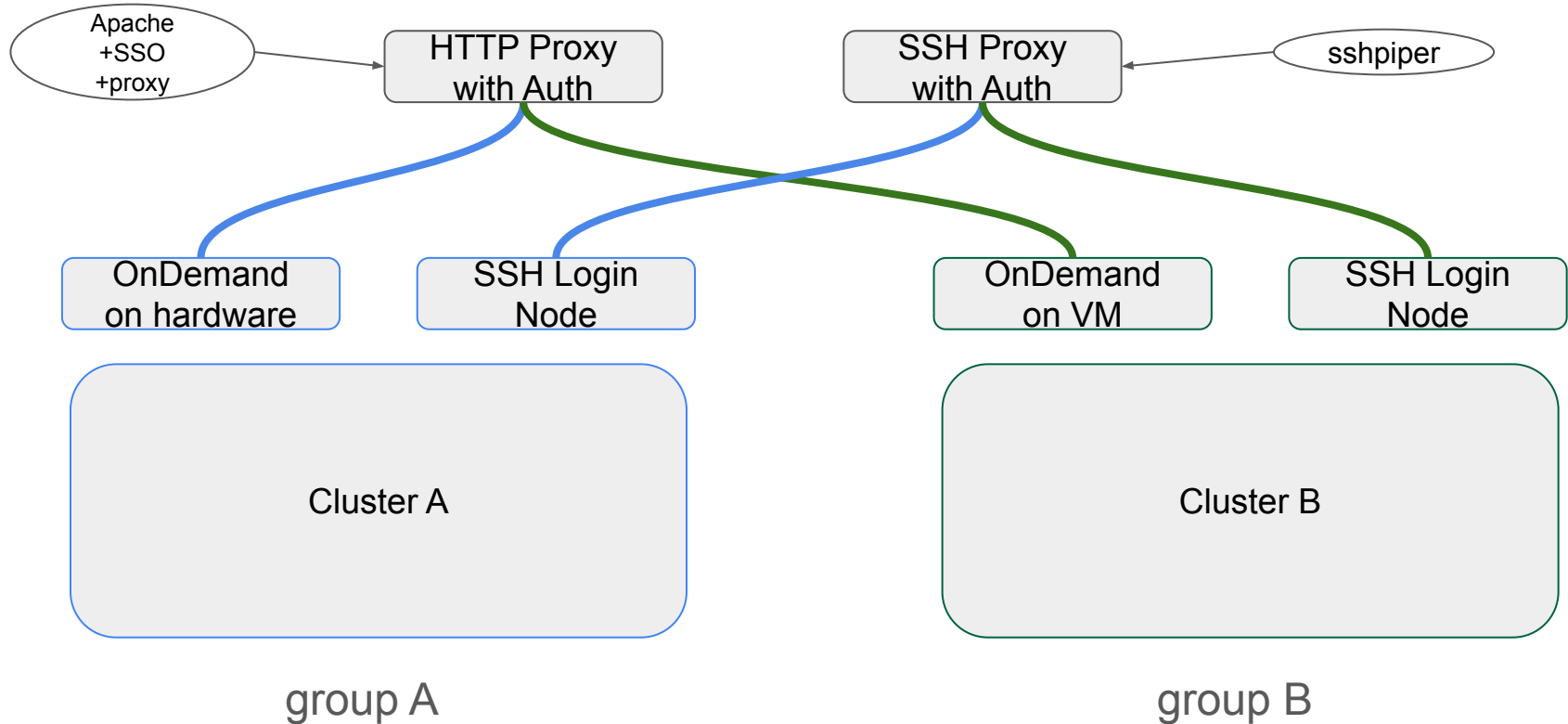
- Stage compute capacity in new data hall connected to GPFS5 caching tier
- Leave existing, production GPFS4 connected HPC undisturbed
- Route user connections for Open onDemand web UI and SSH to their appropriate compute fabric
 - Side A: to-be-migrated users to the original GPFS4 connected fabric
 - Side B: migrated users to the new GPFS5 connected fabric
- Proxy-plane that makes routing decision based on user's migration state
 - Web proxy: use Apache front-end with SSO to route user based on their status
 - SSH proxy: use ssh-piper to route user based on their status

Approach for moving a live HPC system...

- Stage compute capacity in new data hall connected to GPFS5 caching tier
- Leave existing, production GPFS4 connected HPC undisturbed
- Route user connections for Open onDemand web UI and SSH to their appropriate compute fabric
 - Side A: to-be-migrated users to the original GPFS4 connected fabric
 - Side B: migrated users to the new GPFS5 connected fabric
- Proxy-plane that makes routing decision based on user's migration state
 - Web proxy: use Apache front-end with SSO to route user based on their status
 - SSH proxy: use ssh-piper to route user based on their status

.. btw, the proxy plane is the same tooling needed for A/B testing...

What it looks like



Challenges with this much change

- Build process is complex
- Product is the sum of many pieces of software
- Track the changes from the upstream
- Regularly deploy an integrated whole

Keeping track of all the details feels impossible

Need reliable way to inject changes

- The hard way: deploy services to bare-metal
 - Final mile is different from all your prior deploys in virtual environments
 - Can leverage software-defined infrastructure tooling
 - Hard truth: many procedures require network configuration that is out-of-scope for infrastructure tooling
 - Not to mention you need surplus hardware in dev to test scripted deploy
- The easy way: adopt cloud development models and cloud platforms
 - Now infrastructure looks the same, in dev *and* prod
 - Easily scripted and tested with many deploy frameworks
 - Promotes adoption of modern tooling

CI/CD Tooling

- GitLab supports CI/CD workflows out-of-the-box
- Already using Ansible for system configurations
 - CRI_XCBC Ansible code base for much of HPC platform (needs refresh)
 - Bright Cluster-on-Demand (CoD) in Heat
 - Proxy nodes are lightweight
 - Login SSH and Open onDemand node require cluster-specific hooks
- Use Packer to build images
- Use Terraform to deploy...eventually
 - For now it's scripted OpenStack CLI
 - Note: do have TF in other more isolated environments

CI/CD Pipelines

Variables

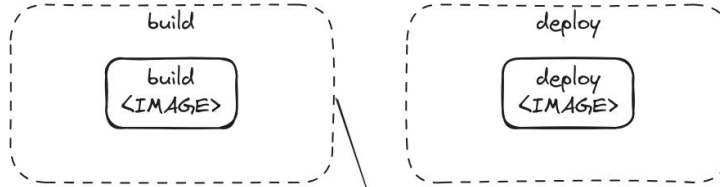
IMAGE:

- ood
- login
- ood-proxy
- ssh-proxy

ENV:

- dev
- test
- staging
- prod

One build per pipeline

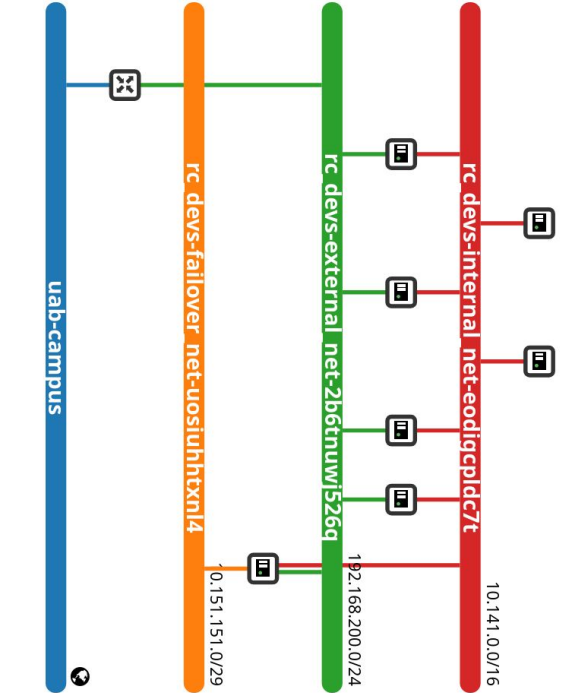
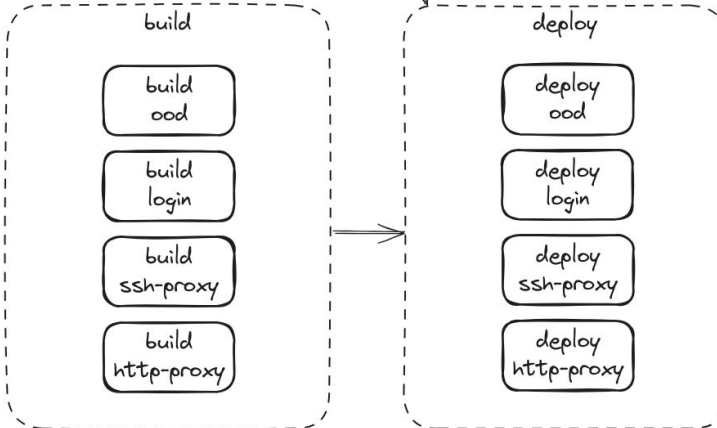


Variables

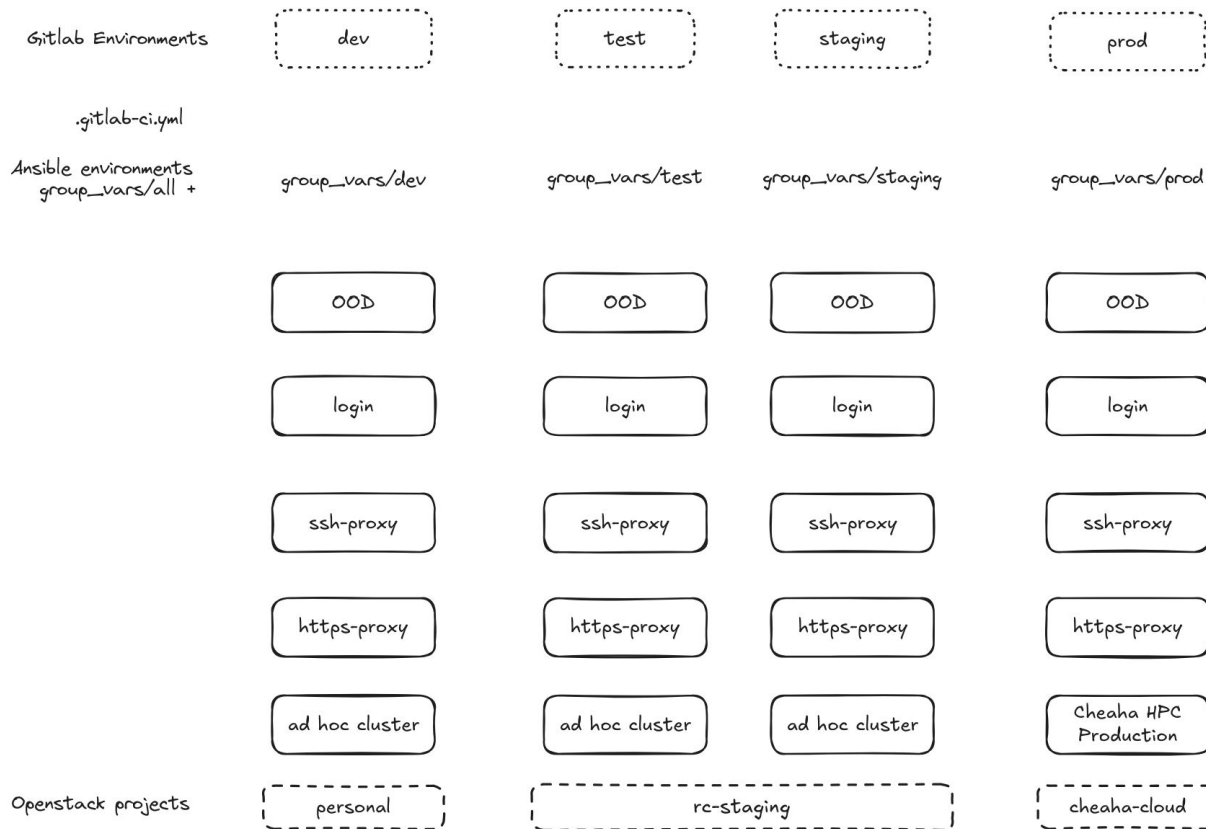
ENV:

- dev
- test
- staging
- prod

One pipeline build all

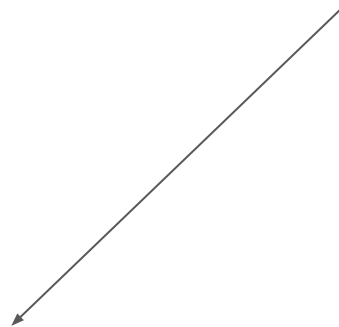


Isolated environments for dev, test, staging and prod



Cluster_hooks

- `ldap_config`
- `nfs_config` (mounts)
- `Slurm_config`
- `dns_config`



Insights and next steps...

- CI/CD empowers work at scale
- Know when feature dev breaks platform
- Baking CD/CD into infrastructure enables others to do it too
- More construction == more experimentation
- Patterns impose order
- Building levers of control over infrastructure
- Next phase:

Give science engagement team control over infrastructure

demo-ish

File Edit View History Bookmarks Profiles Tools Help

UAB CAS - Central Authentic: x +

padlock.idm.uab.edu Search

UAB The University of Alabama at Birmingham.

UAB Central Authentication System

Enter your BlazerID and Password:

BlazerID:

Password:

Log In

Having trouble logging in?

Visit [BlazerID Central](#) for help or to reset your password.

Contact AskIT at 205-996-5555 if you have any problems using this system or your BlazerID.

Security Notice

For security reasons, quit your web browser when finished accessing services that require authentication. Be wary of any program or web page that asks for your BlazerID and password. Legitimate UAB web pages prompting for your BlazerID and password should have addresses that begin with https:// (not http:) and contain "uab.edu". Also, your browser should visually indicate that you are accessing a secure page, usually a padlock icon.

This system is available only for authorized purposes by authorized users. Use for any other purpose may result in disciplinary action or criminal prosecution against the user.

[Disclaimer](#) · [Web Pages Policy](#) · [Contact](#) · © 2025 UAB

gpfs4

File Edit View History Bookmarks Profiles Tools Help

UAB CAS - Central Authentic: x +

padlock.idm.uab.edu/cas/login Private browsing Search

UAB | The University of Alabama at Birmingham

UAB Central Authentication System

Enter your BlazerID and Password:

BlazerID:

Password:

Log In

Having trouble logging in?

Visit [BlazerID Central](#) for help or to reset your password.

Contact AskIT at 205-996-5555 if you have any problems using this system or your BlazerID.

Security Notice

For security reasons, quit your web browser when finished accessing services that require authentication. Be wary of any program or web page that asks for your BlazerID and password. Legitimate UAB web pages prompting for your BlazerID and password should have addresses that begin with https:// (not http:) and contain "uab.edu". Also, your browser should visually indicate that you are accessing a secure page, usually a padlock icon.

This system is available only for authorized purposes by authorized users. Use for any other purpose may result in disciplinary action or criminal prosecution against the user.

[Disclaimer](#) · [Web Pages Policy](#) · [Contact](#) · © 2025 UAB

gpfs5

File Edit View History Bookmarks Profiles Tools Help

Dashboard x +

rc.uab.edu/pun/sys/d

UAB Research Computing Files Jobs Clusters Interactive Apps



OnDemand provides an integrated, single access point for all of your HPC resources.

Message of the Day

Cheaha is a high-performance computing (HPC) platform available to all researchers in the UAB research community, and is operated by the UAB IT Research Computing (UABRC) team.

If you encounter any issues while using Open OnDemand, please email support@listserv.uab.edu with OOD included in the subject line.

For more information on Cheaha and the tools available to support research please review the documentation at <https://docs.rc.uab.edu/>. For more information on using Open OnDemand, please see https://docs.rc.uab.edu/cheaha/open_ondemand/ood_main/.

----- Available Slurm Partitions -----

Partition	Nodes	Nodes Per Researcher	Time Limit	Priority Tier
CPU				
interactive	52	1	2 hours	20

powered by
OPEN OnDemand

OnDemand version: v1.6.25 | Dashboard version:
v1.35.3

gpfs4

File Edit View History Bookmarks Profiles Tools Help

Dashboard - UAB Research x +

rc.uab.edu/pun/sys/dashboard

UAB Research Computing Files Jobs Clusters Interactive Apps



OnDemand provides an integrated, single access point for all of your HPC resources.

Message of the Day

Cheaha is a high-performance computing (HPC) platform available to all researchers in the UAB research community, and is operated by the UAB IT Research Computing (UABRC) team.

If you encounter any issues while using Open OnDemand, please email support@listserv.uab.edu with OOD included in the subject line.

For more information on Cheaha and the tools available to support research please review the documentation at <https://docs.rc.uab.edu/>. For more information on using Open OnDemand, please see https://docs.rc.uab.edu/cheaha/open_ondemand/ood_main/.

----- Available Slurm Partitions -----

Partition	Nodes	Nodes Per Researcher	Time Limit	Priority Tier
CPU				
interactive	52	1	2 hours	20
express	52	UNLIMITED	2 hours	20
short	52	44	12 hours	16
medium	52	44	2 days, 2 hours	12

powered by
OPEN OnDemand

OnDemand version: 2.0.32

gpfs5

FileEditViewHistoryBookmarksProfilesToolsHelp

Dashboard

rc.uab.edu/pun/sys/d

Search

UAB Research ComputingFilesJobsClustersInteractive Apps

CryoEM

RELION

Desktops

HPC Desktop

GUIs

ANSYS

IGV

MATLAB

SAS

Servers

Jupyter Notebook

JupyterLab

RStudio Server



cheaha

supercomputer

OnDemand provides an integrated, single access point to all research resources.

Message of the Day

Cheaha is a high-performance computing (HPC) platform available to all researchers in the UAB research community, and is operated by the UAB IT Research Computing (UABRC) team.

If you encounter any issues while using Open OnDemand, please email support@listserv.uab.edu with OOD included in the subject line.

For more information on Cheaha and the tools available to support research please review the documentation at <https://docs.rc.uab.edu/>. For more information on using Open OnDemand, please see https://docs.rc.uab.edu/cheaha/open_ondemand/ood_main/.

----- Available Slurm Partitions -----

Partition	Nodes	Nodes Per Researcher	Time Limit	Priority Tier
CPU				
interactive	52	1	2 hours	20
express	52	UNLIMITED	2 hours	20
short	52	44	12 hours	16

gpfs4

FileEditViewHistoryBookmarksProfilesToolsHelp

Dashboard - UAB Research

rc.uab.edu/pun/sys/dashboard

Search

UAB Research ComputingFilesJobsClustersInteractive Apps

CryoEM

RELION

Desktops

HPC Desktop

GUIs

IGV

MATLAB

SAS

Servers

Jupyter Notebook

JupyterLab

RStudio Server



cheaha

supercomputer

OnDemand provides an integrated, single access point to all research resources.

Message of the Day

Cheaha is a high-performance computing (HPC) platform available to all researchers in the UAB research community, and is operated by the UAB IT Research Computing (UABRC) team.

If you encounter any issues while using Open OnDemand, please email support@listserv.uab.edu with OOD included in the subject line.

For more information on Cheaha and the tools available to support research please review the documentation at <https://docs.rc.uab.edu/>. For more information on using Open OnDemand, please see https://docs.rc.uab.edu/cheaha/open_ondemand/ood_main/.

----- Available Slurm Partitions -----

Partition	Nodes	Nodes Per Researcher	Time Limit	Priority Tier
CPU				
interactive	52	1	2 hours	20
express	52	UNLIMITED	2 hours	20
short	52	44	12 hours	16
medium	52	44	2 days, 2 hours	12
long	52	5	6 days, 6 hours	8
		5	6 days, 6 hours	8

docs.rc.uab.edu/cheaha/open_ondemand/ood_main/

gpfs5

```
File Edit View History Bookmarks Profiles Tools Help
Dashboard x jpr@login004:~ x +
rc.uab.edu/pun/sys/shell/
Cheaha is an analysis and computing resource available to the UAB research community and operated by the UAB IT Research Computing group.
Use of this resource is governed by the UAB Acceptable Use Policy for Computer and Network Resources. Please review these policies on-line:
https://www.uab.edu/policies/content/Pages/UAB-IT-POL-0000004.html
For more information on Cheaha and the tools available to support research please review the documentation:
https://docs.rc.uab.edu/
----- Environment Modules -----
'module avail' - show available modules
'module add <module>' - adds a module to your environment for this session
'module initadd <module>' - configure module to be loaded at every login
----- Available Slurm Partitions -----
| Partition | Nodes | Nodes Per Researcher | Time Limit | Priority Tier |
|-----|-----|-----|-----|-----|
| **CPU** | | | | |
| interactive | 52 | 1 | 2 hours | 20 |
| express | 52 | UNLIMITED | 2 hours | 20 |
| short | 52 | 44 | 12 hours | 16 |
| medium | 52 | 44 | 2 days, 2 hours | 12 |
| long | 52 | 5 | 6 days, 6 hours | 8 |
| intel-dcb | 21 | 5 | 6 days, 6 hours | 8 |
| amd-hdr100 | 33 | 5 | 6 days, 6 hours | 8 |
| **GPU** | | | | |
| amperenodes | 20 | 2 | 12 hours | 16 |
| amperenodes-medium | 10 | 2 | 2 days, 0 hours | 16 |
| pascalnodes | 18 | UNLIMITED | 12 hours | 16 |
| pascalnodes-medium | 7 | UNLIMITED | 2 days, 0 hours | 15 |
| **Large mem** | | | | |
| largemem | 14 | 10 | 2 days, 2 hours | 6 |
| largemem-long | 5 | 10 | 6 days, 6 hours | 6 |
+-----+-----+-----+-----+-----+
Disk Quota Report
/data/user/jpr + /home/jpr : 4699.36GB of 5476.08GB
/scratch/jpr : .77GB - Please keep scratch clean!
[jpr@login004 ~]$
```

gpfs4

```
File Edit View History Bookmarks Profiles Tools Help
Dashboard - UAB Research x jprorama@gmail.com@login x +
rc.uab.edu/pun/sys/shell/ssh/login006
Host: login006 Themes: Default
**CPU** | | | |
interactive | 52 | 1 | 2 hours | 20 |
express | 52 | UNLIMITED | 2 hours | 20 |
short | 52 | 44 | 12 hours | 16 |
medium | 52 | 44 | 2 days, 2 hours | 12 |
long | 52 | 5 | 6 days, 6 hours | 8 |
intel-dcb | 21 | 5 | 6 days, 6 hours | 8 |
amd-hdr100 | 33 | 5 | 6 days, 6 hours | 8 |
| | | |
**GPU** | | | |
amperenodes | 20 | 2 | 12 hours | 16 |
amperenodes-medium | 10 | 2 | 2 days, 0 hours | 16 |
pascalnodes | 18 | UNLIMITED | 12 hours | 16 |
pascalnodes-medium | 7 | UNLIMITED | 2 days, 0 hours | 15 |
| | | |
**Large mem** | | | |
largemem | 14 | 10 | 2 days, 2 hours | 6 |
largemem-long | 5 | 10 | 6 days, 6 hours | 6 |
+-----+-----+-----+-----+
grep: /data/.mmrepquota_user.log: No such file or directory
grep: /scratch/.mmrepquota_scratch.log: No such file or directory
----- Disk Quota Report -----
/data/user/jprorama@gmail.com + /home/jprorama@gmail.com : 0GB of 5000GB
/scratch/jprorama@gmail.com : 0GB - Please keep scratch cl
ean!
[jprorama@gmail.com@login006 ~]$
```

gpfs5

```
jpr@laptop: ~  
(base) jpr@laptop:~$ ssh -l jpr cheaha.rc.uab.edu
```

gpfs4

```
jpr@laptop: ~  
(base) jpr@laptop:~$ ssh -l jprorama@gmail.com cheaha.rc.uab.edu
```

gpfs5

```
jpr@login004:~
hours | 16 |
| medium | 52 | 44 | 2 days, 2
hours | 12 |
| long | 52 | 5 | 6 days, 6
hours | 8 |
| intel-dcb | 21 | 5 | 6 days, 6
hours | 8 |
| amd-hdr100 | 33 | 5 | 6 days, 6
hours | 8 |
|
| **GPU** |
| amperenodes | 20 | 2 | 12
hours | 16 |
| amperenodes-medium | 10 | 2 | 2 days, 0
hours | 16 |
| pascalnodes | 18 | UNLIMITED | 12
hours | 16 |
| pascalnodes-medium | 7 | UNLIMITED | 2 days, 0
hours | 15 |
|
| **Large mem** |
| largemem | 14 | 10 | 2 days, 2
hours | 6 |
| largemem-long | 5 | 10 | 6 days, 6
hours | 6 |

+-----+-----+-----+-----+
+-----+
Disk Quota Report
+-----+
/data/user/jpr + /home/jpr : 4699.36GB of 5476.08GB
/scratch/jpr : .77GB - Please keep scratch clean!
[jpr@login004 ~]$
```

gpfs4

```
jprorama@gmail.com@login006:~
| long | 52 | 5 | 6 days, 6
hours | 8 |
| intel-dcb | 21 | 5 | 6 days, 6
hours | 8 |
| amd-hdr100 | 33 | 5 | 6 days, 6
hours | 8 |
|
| **GPU** |
| amperenodes | 20 | 2 | 12
hours | 16 |
| amperenodes-medium | 10 | 2 | 2 days, 0
hours | 16 |
| pascalnodes | 18 | UNLIMITED | 12
hours | 16 |
| pascalnodes-medium | 7 | UNLIMITED | 2 days, 0
hours | 15 |
|
| **Large mem** |
| largemem | 14 | 10 | 2 days, 2
hours | 6 |
| largemem-long | 5 | 10 | 6 days, 6
hours | 6 |

+-----+-----+-----+-----+
+-----+
grep: /data/.mmrepquota_user.log: No such file or directory
grep: /scratch/.mmrepquota_scratch.log: No such file or directory
+-----+
Disk Quota Report
+-----+
/data/user/jprorama@gmail.com + /home/jprorama@gmail.com : 0GB of 5000GB
/scratch/jprorama@gmail.com : 0GB -
Please keep scratch clean!
[jprorama@gmail.com@login006 ~]$
```

gpfs5